

THEORY AND METHODS
A BIPLLOT BASED APPROACH TO
DISCRIMINANT ANALYSIS WITH
CATEGORICAL VARIABLES IN THE
PRESENCE OF REVERSALS

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Summary: Gower and Hand (1996) provide a unified biplot theory in which biplots are seen as multivariate analogues of ordinary scatterplots. In this paper a perspective is given on a biplot based approach to problems in discrimination. Various types of biplots are introduced for use when performing discriminant analysis in different situations. The focus is on discriminant analysis with categorical predictors. In particular it is shown how biplot based procedures can efficiently overcome the well-known problems that occur with binary predictors in the presence of reversals in log likelihood ratios, i.e. where the log likelihood ratio depends only on the number of positive coded elements of a random vector but it is not a monotone function of this number.

It is argued that generalised biplot methodology allows the inclusion of categorical variables in a natural way by defining a distance matrix where both continuous and categorical variables contribute to the inter-point distances. A proposal is made where such distance matrices together with non-parametric procedures such as MARS and BRUTO are used as input to the non-linear

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biplot method to perform discriminant analysis. The potential of this proposal to address the reversal problem is investigated by means of a simulation study. Conditions favourable to linear discriminant analysis as well as non-linear discriminant analysis are considered. Examples of the associated biplots are provided.

1. Introduction

Biplots have many applications in practice. In particular, Gardner (2001) provides several extensions of biplot methodology for application in discriminant analysis and classification problems. These extensions address a wide variety of situations arising in discriminant analysis. It follows that combining the powerful graphical information offered by biplots with the ability of the underlying biplot methodology to actually perform the discriminant analysis or classification itself, provides researchers in these fields with a powerful research instrument.

In this paper the focus is on discriminant analysis with categorical predictor variables and in particular the ease of dealing with these categorical predictors by formulating discriminant analysis in terms of biplot methodology. Furthermore, since it is known that categorical predictors can cause problems in certain discrimination situations, in particular where so-called reversals are present, it will be shown how these problems can be overcome efficiently by using biplot methodology. Binary categorical predictors are common in practice. When such predictors are coded as zero or one, for example, the log likelihood ratio forming the basis of a classification procedure can be considered as a function of the number of ones. In some practical applications this function is not monotonic: the log likelihood function, for example, will

show at first an increase as the number of ones increases; then it decreases before showing once again an increase as the number of ones increases further. A reversal is said to occur in applications where the log likelihood function is characterised by such behaviour. Before turning attention to the above matters a brief perspective is given on biplot methodology in general and on a biplot based approach to discrimination techniques in particular.

Gardner and Le Roux (2003) propose a biplot based approach to discriminant analysis with categorical variables. They provide an introduction to biplot methodology emphasising the perspective of Gower and Hand (1996) according to which biplots, originally introduced by Gabriel (1971), are regarded as multivariate extensions of ordinary scatterplots. Gower and Hand's perspective offers a unifying theory encompassing principal component analysis (PCA) biplots, canonical variate analysis (CVA) biplots, non-linear biplots and generalised biplots. The PCA biplot may be regarded as a simple eigenvalue problem and is based upon Pythagorean distance (also known as Euclidean distance). A CVA biplot results from solving a two-sided eigenvalue problem and is based upon Mahalanobis distance. Now, suppose that a multidimensional scaling (MDS) procedure is performed on a distance matrix originating from any non-Euclidean distance metric and that an associated biplot is sought. Gower and Hand (1996) consider the case where the distance matrix is embeddable in a Euclidean space as well as the case where it is not. They discuss construction of the non-linear biplot in such cases. In particular they show that explicit forms for both interpolative and predictive biplot axes can be derived for Euclidean embeddable distances satisfying an additivity property and where the method of MDS is principal coordinate analysis. These axes are in the form of non-linear trajectories. The methodology can be

extended further to provide for categorical variables leading to the construction of a generalised biplot.

According to the Gower and Hand philosophy the principal axes found by solving the respective eigenvalue problems form the scaffolding for representing the multidimensional sample points but are themselves not shown in the biplot. The aim is to represent the sample points in such a way that the distances between them are preserved. Instead of plotting the principal axes the variables are represented as axes (continuous variables) or category level points (categorical variables). Gower and Hand (1996) introduce the terms *interpolation* for the placement of a (new) sample point on the biplot display and *prediction* for inferring the values of the original variables for any point in the display. Therefore two sets of biplot axes are needed: interpolation axes and prediction axes. It can be shown that in most cases these two sets of axes differ and Gardner (2001) suggests biplots to be equipped with prediction axes as it is natural to use axes for prediction purposes. Interpolation is taken care of by a suitable computer programme.

In this paper we aim to show the relative merits of several biplot based methods for performing discriminant analysis when categorical predictors are present. Specifically, the classification performance of these methods when a reversal occurs when dealing with binary predictors will be highlighted. Keeping these aims in mind the remainder of the paper is organised as follows: the next section provides a brief overview of discrimination from the perspective of biplot methodology. This is followed by a section where it is argued that biplot methodology naturally allows inclusion of categorical variables by operating primarily on a distance matrix where both continuous and categorical variables contribute to the inter-sample distances. Since the focus is on binary variables in the presence of a reversal, a separate section is

devoted to this topic before proposing biplot based classification procedures for dealing with reversals. The results of two simulation studies to investigate the classification performance of the proposed methods are reported. Properties of different error rates associated with the proposed methods are discussed and the paper concludes with a discussion of the findings.

2. Perspective on a biplot based approach to discrimination

The CVA biplot is the simplest application of biplot methodology in discriminant analysis and can be regarded as a graphical version of linear discriminant analysis (LDA). In the CVA biplot the Mahalanobis distances between the class means in the original observation space are converted to ordinary Pythagorean distances in the canonical space. Gardner (2001) interpolates all sample points onto the biplot of class means and discusses several proposals for describing, displaying and quantifying multidimensional class separation and overlap.

The non-linear biplot, discussed in detail in Gower and Harding (1988) and Gower and Hand (1996) provides a general methodology for constructing biplots. Both PCA and CVA biplots are special cases of the non-linear biplot and Gardner (2001) shows how to use this non-linear biplot methodology for performing LDA. PCA biplots are obtained as a special case of the non-linear biplot by applying the non-linear biplot method to an inter-sample distance matrix derived from Pythagorean distances and similarly the CVA biplot from such a distance matrix derived from Mahalanobis distances. In the case of a CVA biplot a sample is classified according to its nearest Pythagorean distance from a class mean in the biplot space. Although the biplot display space is two-dimensional the biplot methodology can be used to perform the actual

discriminant analysis in any subspace of the canonical space which is of dimension $K = \min(p, J - 1)$ where p denotes the number of predictors and J the number of classes.

The algorithm provided by Gower and Hand (1996) for displaying category level points associated with categorical variables in generalised biplots is utilised in the non-linear biplot for equipping the latter with classification regions. In addition to providing a visual display of the classification of the samples this biplot methodology allows for a reduction in the dimension of the space in which the actual discriminant analysis takes place. This is important since Hastie, Tibshirani and Buja (1994) comment that a reduced space can be more stable and therefore better classification performance can be expected in some instances of dimension reduction. The non-linear biplot method allows for an alternative classification scheme: classify samples to their nearest approximated canonical mean in the biplot space. A series of possible classification schemes of dimension 2, 3, ..., K , respectively can thus be obtained.

When the non-linear biplot methodology is applied to an inter-sample distance matrix derived from Pythagorean distances, the result is a PCA biplot. Similar to the construction of classification schemes outlined above for CVA biplots such classification schemes can also be constructed for this PCA biplot but the criterion is now nearest Mahalanobis distance to class means. Although this might seem disadvantageous the advantage of using the above PCA biplot procedure is that it allows generalisations to other discrimination procedures.

It is easy to adapt the above PCA classification procedure when the assumption of equal within-class covariance matrices is violated: for each point in the biplot space, instead of using a common within-class covariance matrix, calculate the Mahalanobis distance of that point from each of the class means

using the within-class covariance matrix of each class in turn and classify the point according to the minimum value obtained. However, it is intuitively more appealing if a quadratic discriminant analysis (QDA) biplot can be constructed where distances between class means are Pythagorean. This can be approached by attempting to maximise simultaneously J ratios of between to within-class variance ratios. According to Graybill (1983) a solution to this problem exists if and only if the J within-class covariance matrices pairwise commute. This is generally not the case and therefore the FG-algorithm of Flury and Gautschi (1986) is used to simultaneously bring the J matrices as close to diagonality as possible. The matrix thus obtained is then used to form the scaffolding for constructing a QDA biplot.

The large number of parameters that has to be estimated in QDA can in practice result in over-parameterisation inducing a loss of power. This leads to LDA sometimes outperforming QDA even though the assumption of equal within-class covariance matrices is violated (Flury, Boukai and Flury, 1997; McLachlan, 1992). Hastie, Tibshirani and Buja (1994) propose flexible discriminant analysis (FDA) as a compromise solution that is less rigid than LDA but involves fewer parameters to be estimated than QDA. This aim is achieved by utilising the relationship between LDA and multiple regression and replacing the conventional multivariate regression procedure with a non-parametric multivariate regression procedure. Essentially in FDA observations are transformed to a 'canonical space' where the transformation involves the fitted values resulting from a non-parametric multivariate regression procedure. Once these FDA observations are available, Pythagorean distances between them can be calculated to form a distance matrix for input to the non-linear biplot method. The resulting biplot is termed an FDA biplot (Gardner, 2001). Since distances between class means in the FDA biplot space are Pythagorean,

classification regions can be constructed easily. Gardner and Le Roux (2005) give a detailed account of the construction of FDA and related biplots. They also provide several examples of such biplots.

3. Discriminant analysis with categorical variables

The generalised biplot defined by Gower (1992) can be seen as a non-linear biplot which makes provision for categorical variables. The only assumption to be met for applying the generalised biplot method is that the inter-sample distances must be additive Euclidean embeddable (Gower and Hand, 1996). Therefore the predictor variables can be arranged such that the first $p_{(1)}$ are continuous and the remaining $p_{(2)}$ are categorical. Let $p = p_{(1)} + p_{(2)}$ then an inter-sample distance matrix

$$D = \sum_{k=1}^{p_{(1)}} D_k + \sum_{k=p_{(1)}+1}^p D_k = D_{(1)} + D_{(2)} \quad (1)$$

can be calculated. Popular choices for the distance measures in (1) are Pythagorean distances for $D_{(1)}$ and for $D_{(2)}$ the extended matching coefficient (EMC) discussed in detail by Gower (1992).

Discriminant analysis using the generalised biplot can be applied to situations with continuous variables only, categorical variables only or a mixture of both. If only continuous variables are used, the data can be represented in the usual way in the matrix X : $n \times p$ with the singular value decomposition $X = U\Lambda V'$. The PCA biplot can be considered a special case of the non-linear biplot using Pythagorean inter-sample distances while the non-linear biplot results as a special case of the generalised biplot applied to continuous variables only. This means that the matrix XV will correspond to the non-linear biplot ordination. Since the matrix V is orthogonal this is only

a rotation of the original samples matrix X . Applying discriminant analysis to the matrix XV is therefore equivalent to using the matrix X .

Care, however, should be taken with the scaling of the data matrix X . As usual X is the centred data matrix. Using Pythagorean inter-sample distance with this centred data matrix X will result in a PCA ordination as described above. However, if categorical variables are present, some scaling of the continuous variables should be considered, e.g. scaled to have unit range or maximum value equal to unity. Applying the Pythagorean inter-sample distance metric to this scaled matrix will result in a slightly different ordination. This slight deviation from the original PCA ordination is, however, a small price to pay compared to the added facility of using categorical predictor variables.

Once the matrix (1) is available it can be used as input to the non-linear biplot method discussed in section 2. Any of the biplot procedures introduced there for classification can then be performed. We will restrict our further discussion to the FDA biplot procedure with the associated non-parametric procedures of MARS and BRUTO (Hastie, Tibshirani and Buja, 1994; Hastie and Tibshirani, 1990). BRUTO is an adaptive method using smoothing splines for estimating an additive model of the form $E(Y) = \sum_{j=0}^p g_j(x_j)$ where the $g_j(\cdot)$ are unknown smooth functions. The above additive model does not provide for interactions between predictors. However, the set of basis functions created by BRUTO for each term in the additive model can be replaced by a product of basis functions to allow the modelling of interactions. In multivariate adaptive regression splines (MARS) expansions in piecewise linear basis functions are used in these products. Both BRUTO and MARS automatically involve a variable selection procedure and both operate on the data locally.

4. Reversal

In the discussion of McLachlan (1992) on the performance of LDA with categorical predictor variables it is concluded that LDA is quite robust for departures from normality if the true log ratio of the class-conditional densities is an (approximately) linear function of the predictor variables. Krzanowski (1977) gives a detailed discussion on the performance of LDA with categorical or mixed predictor variables. In particular, Krzanowski points out that LDA will behave poorly in a two class situation with two binary predictor variables where the values (0, 0) and (1, 1) are more frequently associated with the one class and the values (0, 1) and (1, 0) with the other. Since the true log ratio of the class-conditional densities does not increase monotonically with the number of positive predictor variables, it is said to undergo a reversal (see Figure 3).

Both McLachlan and Krzanowski base their discussions on the simulation results of Moore (1973). Moore's results clarified why Gilbert (1968) referred to by Krzanowski (1977), McLachlan (1992) and Moore (1973) found that LDA performed well in a two class discrimination problem with only binary predictor variables. There was no reversal in Gilbert's simulation studies. An example of a reversal in real data is given by Yerushalmy, Van den Berg, Erhardt and Jacobziner (1965). Birth weight and length of gestation period were used to indicate immaturity in newborn babies. Babies were classified as normal for the combinations low birth weight and short gestation period or high birth weight and long gestation period. Abnormal babies were associated with the remaining combinations of long gestation but low birth weight or short gestation with high birth weight.

In the above example the two predictor variables are positively correlated for normal babies, but negatively correlated for abnormal babies. Therefore

the assumption of equal within-class covariance matrices needed for LDA is not satisfied. McLachlan mentions further that Dillon and Goldstein (1978) demonstrated that reversals can occur even in situations where the within-class covariance matrices are equal. If the within-class covariance matrices differ, QDA could be applied. It was already mentioned in section 2 that LDA sometimes outperforms QDA even when the assumptions for LDA are not satisfied due to the large number of parameters that has to be estimated for QDA. Moore (1973) further concludes that the use of QDA with binary variables is not recommended since it rarely performs as well as LDA.

To investigate the performance of a biplot based discrimination procedure in the case of categorical variables with reversals present, a simulation study similar to that of Moore (1973) is performed. The following definition is necessary:

Definition 1 The Bahadur reparameterisation for a multinomial probability (Bahadur, 1961). Consider a random vector $\underline{X} : p \times 1$ consisting of (not necessarily independent) Bernoulli random variables. A particular realisation of $\underline{X}$, denoted by $\underline{x}' = (x_1, x_2, \dots, x_p)$, is called a response pattern. A response pattern consists of a sequence of 0's and 1's. Let $\pi_j(x)$ denote the probability that response pattern $\underline{x}$ is observed in the j -th population. Bahadur (1961) has shown that the multinomial probabilities $\pi_j(\underline{x})$ can be reparameterised as follows:

$$\begin{aligned} \pi_j(\underline{x}) = & \prod_{i=1}^p p_{ij}^{x_i} (1 - p_{ij})^{1-x_i} \left[1 + \sum_{i < k} r_j(ik) z_{ij} z_{kj} \right. \\ & \left. + \sum_{i < k < l} r_j(ikl) z_{ij} z_{kj} z_{lj} + \dots + r_j(12\dots p) z_{1j} z_{2j} \dots z_{pj} \right] \quad (2) \end{aligned}$$

where $p_{ij} = E_j(X_i)$ is the expectation of X_i in population $j = 1, 2, \dots, J; i = 1, 2, \dots, p$

$$\begin{aligned} Z_{ij} &= \frac{X_i - p_{ij}}{\sqrt{p_{ij}(1 - p_{ij})}} \\ r_j(ik) &= E_j(Z_{ij}Z_{kj}) \\ r_j(ikl) &= E_j(Z_{ij}Z_{kj}Z_{lj}) \\ &\vdots \\ r_j(12\dots p) &= E_j(Z_{1j}Z_{2j}\dots Z_{pj}). \end{aligned}$$

In definition 1 Z_{ij} is a random variable with mean 0 and variance 1. The covariance matrix of $\underline{Z}_j$ is the correlation matrix

$$P_j = \begin{bmatrix} 1 & r_j(12) & \dots & r_j(1p) \\ r_j(12) & 1 & \dots & r_j(2p) \\ \dots & \dots & \dots & \dots \\ r_j(1p) & r_j(2p) & \dots & 1 \end{bmatrix} \quad (3)$$

where the $r_j(ik)$ satisfies $-1 \leq r_j(ik) \leq 1$ such that (3) is positive definite, e.g. for the equicorrelated model $r_j(ik) > -1/(p-1)$ (Graybill, 1983).

A first order approximation to the Bahadur reparameterisation results when all r_j are zero; a second order approximation when not all $r_j(ik)$ are zero, but all $r_j(ikl)$ and higher are zero. Higher-order approximations are defined similarly.

Moore (1973) used the second-order approximation to the Bahadur reparameterisation of the multinomial distribution. According to this approximation the probability of observing the response vector (response pattern) $\underline{x}$, a p -dimensional observation of a vector of p Bernoulli variables, in population j is given by

$$\pi_j(\underline{x}) = \prod_{i=1}^p p_{ij}^{x_i} (1 - p_{ij})^{1-x_i} \left\{ 1 + \sum_{i < k} r_j(ik) z_{ij} z_{kj} \right\}. \quad (4)$$

5. A biplot based classification procedure for dealing with reversals

Gardner and Le Roux (2003) report the results of applying biplot based discriminant analysis procedures to simulated data similar to that of Yerushalmy, et al. (1965). They demonstrate how to include categorical variables in the discrimination process by obtaining an ordination with the generalised biplot method. After obtaining this ordination Gardner and Le Roux resolved the problem of classification in the presence of reversals by changing the distance metric in the non-linear biplot classification procedure.

This paper focuses on a simulation study to compare the performance of the biplot based method with the results reported by Moore (1973). In Moore's study nineteen pairs of populations were formed by specifying different values for the parameters in the Bahadur reparameterisation. In each case six variables were used. These population pairs can be divided into three groups, one where all correlations $r_j(ik) = 0$, the second with a single non-zero correlation term and the final group included those with all correlations positive. Since this paper aims at comparing the effectiveness of LDA and FDA biplot classifications with binary variables in the presence of a reversal as well as in situations that are favourable for LDA, only four of Moore's configurations are considered: Configurations 2 and 5 (no reversal; LDA effective; populations further apart in Configuration 5 than in 2); Configurations 3 and 6 (reversals; LDA ineffective; populations further apart in Configuration 6 than in 3). For these configurations the parameters of the respective population pairs were specified as follows:

Configuration 2:

$p_{i1} = (0.2, 0.2, 0.2, 0.2, 0.2, 0.2)$ and all correlations zero except $r_1(13) = 0.5$

$p_{i2} = (0.4, 0.4, 0.4, 0.4, 0.4, 0.4)$ and all correlations zero except $r_2(13) = 0.5$.

Configuration 5:

$p_{i1} = (0.2, 0.2, 0.2, 0.2, 0.2, 0.2)$ and all correlations zero except $r_1(13) = 0.5$

$p_{i2} = (0.6, 0.6, 0.6, 0.6, 0.6, 0.6)$ and all correlations zero except $r_2(13) = 0.5$.

Configuration 3:

$p_{i1} = (0.2, 0.2, 0.2, 0.2, 0.2, 0.2)$ and all $r_1(ik) = 0.3$

$p_{i2} = (0.4, 0.4, 0.4, 0.4, 0.4, 0.4)$ and all $r_2(ik) = 0.3$.

Configuration 6:

$p_{i1} = (0.2, 0.2, 0.2, 0.2, 0.2, 0.2)$ and all $r_1(ik) = 0.3$

$p_{i2} = (0.6, 0.6, 0.6, 0.6, 0.6, 0.6)$ and all $r_2(ik) = 0.3$.

The covariance matrix of the response vector $\underline{X}$ in the respective configurations is as follows:

Configurations 2 and 5:

$$\begin{aligned} \sum_{Pop1} &= \begin{bmatrix} 0.16 & 0 & 0.04 & 0 & 0 & 0 \\ 0 & 0.16 & 0 & 0 & 0 & 0 \\ 0.04 & 0 & 0.16 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.16 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.16 \end{bmatrix}; \\ \sum_{Pop2} &= \begin{bmatrix} 0.24 & 0 & 0.12 & 0 & 0 & 0 \\ 0 & 0.24 & 0 & 0 & 0 & 0 \\ 0.12 & 0 & 0.24 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.24 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.24 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.24 \end{bmatrix} \end{aligned}$$

Configurations 3 and 6:

$$\sum_{Pop1} = \begin{bmatrix} 0.160 & 0.048 & 0.048 & 0.048 & 0.040 & 0.048 \\ 0.048 & 0.160 & 0.048 & 0.048 & 0.048 & 0.048 \\ 0.048 & 0.048 & 0.160 & 0.048 & 0.048 & 0.048 \\ 0.048 & 0.048 & 0.048 & 0.160 & 0.048 & 0.048 \\ 0.048 & 0.048 & 0.048 & 0.048 & 0.160 & 0.048 \\ 0.048 & 0.048 & 0.048 & 0.048 & 0.048 & 0.160 \end{bmatrix};$$

$$\sum_{Pop2} = \begin{bmatrix} 0.240 & 0.072 & 0.072 & 0.072 & 0.072 & 0.072 \\ 0.072 & 0.240 & 0.072 & 0.072 & 0.072 & 0.072 \\ 0.072 & 0.072 & 0.240 & 0.072 & 0.072 & 0.072 \\ 0.072 & 0.072 & 0.072 & 0.240 & 0.072 & 0.072 \\ 0.072 & 0.072 & 0.072 & 0.072 & 0.240 & 0.072 \\ 0.072 & 0.072 & 0.072 & 0.072 & 0.072 & 0.240 \end{bmatrix}$$

Moore evaluated LDA and QDA as well as the full multinomial, the first order Bahadur and second order Bahadur procedures. When the true log likelihoods are known, i.e. when $\pi_1(\underline{x})$ and $\pi_2(\underline{x})$ are fully specified the probability of misclassification can be calculated. This probability is defined as the optimum error rate (OER). The actual error rate (AER) results when estimates of the log likelihoods are used for calculating the probability of misclassification. An error rate that does not depend on the form of the parent distributions is the apparent error rate (APER) which is defined as the proportion of misclassifications occurring in the sample used for calculating the particular classification rule. For detailed discussions of the above error rates and their properties see for example Hills (1966) and McLachlan (1992). Moore reports the OER when the full multinomial procedure is used for classification as well as the mean increase in AER over the OER. Based on 50 Monte Carlo samples the error rates in Table 1 are reported by Moore for sample sizes of 50 observations per class and 100 observations per class, as well as the error rates under the assumption that the true $\pi_j(\underline{x})$ are known (indicated by sample size = ∞).

Table 1. OER and mean AER obtained by Moore (1973).

Configuration	$n_1 = n_2 = n$	OER	Mean AER			
			LDA	QDA	Full multinomial	Second order
2	50	0.296	0.324	0.346	0.391	0.320
	100		0.311	0.325	0.368	0.311
	∞		0.296	0.296	0.296	0.302
5	50	0.161	0.177	0.196	0.247	0.170
	100		0.169	0.180	0.208	0.164
	∞		0.161	0.161	0.161	0.161
3	50	0.262	0.380	0.353	0.336	0.403
	100		0.384	0.366	0.317	0.407
	∞		0.405	0.405	0.262	0.405
6	50	0.179	0.290	0.267	0.245	0.322
	100		0.297	0.272	0.227	0.329
	∞		0.335	0.262	0.179	0.335

The simulation study to compare the FDA biplot classification procedure for the above four configurations with Moore's results consists of two parts: in the first part 100 repetitions of each of the above configurations were obtained for sample sizes 50 and 100. Apparent error rates were calculated and compared with results reported by Moore (1973) for the LDA biplot procedure as well as the FDA biplot procedure using MARS for obtaining the non-parametric regression estimates. In the second part in addition to APER, leave-one-out error rates (CV error rates) were also obtained. This was done for at least 120 repetitions in the case of sample size 50. Furthermore the additive BRUTO procedure was also included for obtaining the non-parametric regression estimates as an alternative to MARS that also takes interaction into account.

We first take a look at Configurations 2 and 5, i.e. the configurations favourable to LDA as summarised in Table 2 together with Figure 1 (upper panels).

Inspection of Table 2 and Figure 1 reveals that in the case of Configuration 2 the LDA biplot procedure and the FDA biplot procedure give very similar results. The mean APER for both procedures is slightly better than the OER. The LDA biplot procedure performs marginally better than the FDA biplot procedure when the smaller samples are considered and the FDA biplot procedure is marginally better in the case of the larger samples. The variation of APER in the Monte Carlo repetitions decreases when the sample size is increased.

Table 2. Mean and standard deviation (SD) of APER obtained in 100 repetitions of Moore in the LDA and FDA (MARS) biplot procedures.

Configuration	$n_1 = n_2 = n$	OER	LDA biplot procedure		FDA (MARS) biplot procedure	
			Mean	SD	Mean	SD
2	50	0.296	0.2741	0.0425	0.2849	0.0506
	100		0.2904	0.0348	0.2855	0.0327
5	50	0.161	0.1419	0.0361	0.1329	0.0429
	100		0.1478	0.0221	0.1413	0.0257
3	50	0.262	0.3189	0.0469	0.2538	0.0513
	100		0.3595	0.0359	0.2574	0.0350
6	50	0.179	0.2393	0.0495	0.1433	0.0433
	100		0.2817	0.0332	0.1639	0.0286

Configuration 5 differs from Configuration 2 only in that the expected response patterns of the two populations are further apart. This results in a smaller OER of 0.161. Table 2 and Figure 1 (upper panels) show that the mean APER obtained by the LDA and FDA biplot procedures closely follows this trend. The mean APER in case of both procedures is slightly less than the OER and the variation again decreases with an increase in sample size. The mean APER of the FDA biplot procedure is marginally better than that of the LDA biplot procedure for the smaller as well as the larger samples.

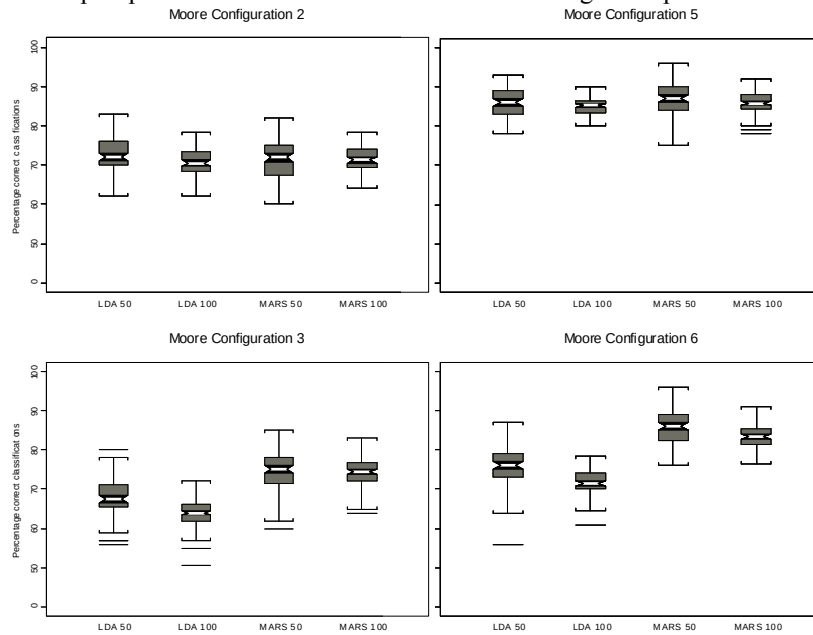


Figure 1. APER Classification performance obtained in 100 repetitions of Moore Configuration 2 (top left panel); Moore Configuration 5 (top right panel); Moore Configuration 3 (bottom left panel) and Moore Configuration 6 (bottom right panel). All four panels scaled to the same vertical range.

In both Configuration 2 and 5 the mean APER of the LDA biplot procedure is marginally smaller than the results reported by Moore (cf. Table 1). This difference may be due to the optimistic bias of the APER.

Comparing in the upper two panels of Figure 1 the notches of LDA and FDA-MARS boxplots in similar circumstances, it is seen that these notches always overlap. Thus we can conclude (cf. McGill, Tukey and Larsen, 1978) that there is no significant difference between the median percentages of correct APER classifications obtained by the LDA biplot procedure and the FDA (MARS) biplot procedure at approximately a 5% significance level.

Now we direct attention to the configurations not favourable to LDA, i.e. Configurations 3 and 6. The results displayed in Table 2 as well as in the bottom two panels of Figure 1 for Configurations 3 and 6 stand in sharp contrast to what were found for Configurations 2 and 5. The LDA biplot procedure leads to a mean APER that is appreciably higher than the OER. This is especially the case when the larger samples are used. However, when the results of the LDA biplot procedure are compared to what was found by Moore (cf. Table 1) it is clear that the application of LDA within the framework of biplot methodology resulted in a better mean APER than the AER reported by Moore (1973). The FDA biplot method, on the other hand, has a mean APER slightly less than the OER. Not only is this appreciably better than obtained by the LDA biplot procedure but it is better than the AER of any of the five procedures considered by Moore. As in the case of Configurations 2 and 5 an increase in the sample size leads to a decrease in Monte Carlo variation.

Configuration 6 differs from Configuration 3 similar to how Configuration 5 differs from Configuration 2. The results obtained with Configuration 6

are similar to those found using Configuration 3 reflecting the increase in $\pi_2(\underline{x}) - \pi_1(\underline{x})$ by a general decrease in the mean APER. The LDA procedure shows a mean APER of approximately 35% higher than the OER in the case of the smaller samples with an even larger difference in the case of the larger samples. Contrary to LDA, the FDA biplot procedures have mean APER less than the OER.

Comparing in the bottom two panels of Figure 1 the notches of LDA and FDA-MARS boxplots in similar circumstances, it is seen that these notches never overlap. Indeed there is even no overlap of the boxes. Thus we can conclude that the median percentages of correct APER classifications obtained by the FDA-MARS biplot procedure are statistically significantly higher than those obtained by the LDA biplot procedure.

The second part of the simulation study was undertaken to address the following issues: since it is well known that the APER is optimistically biased, what will happen if a cross-validation error rate (CV-ER) is computed for comparison purposes; would there be any change in the FDA biplot classification results if another method is used for obtaining the necessary non-parametric regression estimates; and lastly approximately how many Monte Carlo replications are adequate? Since in the first part of the study, apart from smaller standard errors, very similar results were obtained for sample sizes 50 and 100, only the smaller sample size was considered in the second phase. The BRUTO procedure was included as an alternative method for obtaining the non-parametric regression estimates necessary for the FDA biplot classification procedure.

We summarise the results obtained in the second part of the study in Table 3 together with the boxplots given in Figure 2.

Table 3. Mean and standard deviation of APER and CV Error Rate obtained in 121 repetitions of Moore Configuration 2; 127 repetitions of Configuration 5; 121 repetitions of Configuration 3 and 122 repetitions of Configuration 6. In all cases $n_1 = n_2 = 50$.

Conf	OER	LDA biplot procedure				FDA (MARS) biplot procedure				FDA (BRUTO) biplot procedure			
		APER		CV-ER		APER		CV-ER		APER		CV-ER	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
2	0.296	0.276	0.038	0.326	0.054	0.277	0.043	0.357	0.064	0.290	0.043	0.341	0.067
5	0.161	0.147	0.034	0.184	0.040	0.143	0.039	0.199	0.046	0.156	0.036	0.182	0.042
3	0.262	0.323	0.046	0.368	0.051	0.250	0.049	0.314	0.062	0.297	0.078	0.334	0.069
6	0.179	0.246	0.052	0.289	0.057	0.144	0.044	0.221	0.050	0.207	0.087	0.259	0.056

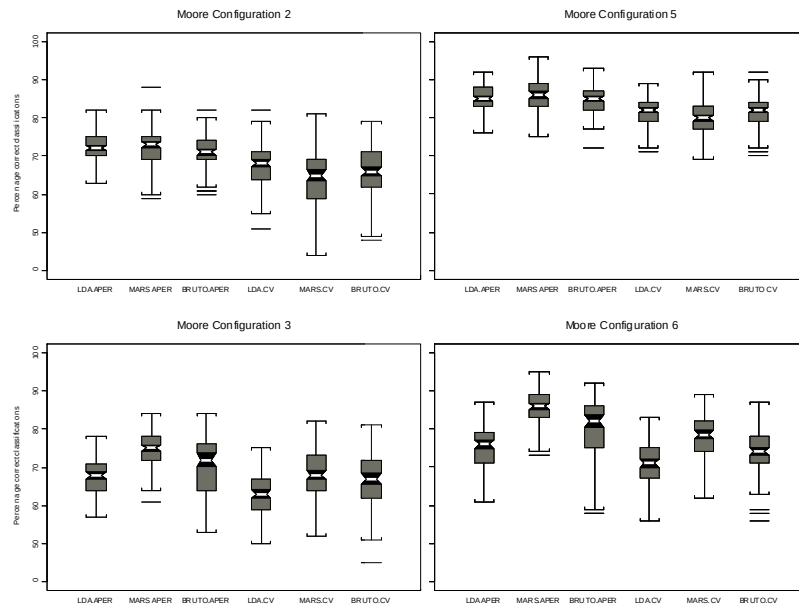


Figure 2. APER and CV Classification performance obtained in 121 repetitions of Moore Configuration 2 (top left panel); 127 repetitions of Configuration 5 (top right panel); 121 repetitions of Configuration 3 (bottom left panel) and 122 repetitions of Configuration 6 (bottom right panel). All four panels scaled to the same vertical range.

The results displayed in Table 3 and Figure 2 confirm the finding in the first part of the study that there is no significant difference in the performance of the LDA and FDA biplot classification procedures in the absence of reversals (Configurations 2 and 5). However, important additional conclusions can be drawn: although the corresponding CV error rates are as expected higher than the corresponding APERs, this is particularly true of FDA with MARS. Comparing the boxplots in Figure 2 (upper panels), the overlap in the notches of the three procedures in the case of APER is clear but in the case of CV there

are indications that the FDA biplot procedure based upon BRUTO outperforms the procedure where MARS is used.

Lastly we summarise the results obtained in the second part of the study for Configurations 3 and 6 in Table 3 together with the boxplots given in the bottom panels of Figure 2. Perusal of these results confirms what was found in the earlier part of the study viz. that the FDA biplot classification procedure performs significantly better than the LDA biplot procedure (and at least as well as the other procedures considered by Moore where such comparisons are possible). As expected, error rates based upon CV are larger than APER for comparable cases. Comparing the notches of comparable boxplots in Figure 2 (bottom panels), confirms that the FDA biplot classification procedure performs statistically significantly better than LDA. However, compared to the FDA biplot procedure with BRUTO the FDA biplot procedure using MARS performs better in all comparable cases with reversals present. This is particularly true in the case of Configuration 6 where the relevant notches are far apart. This is in contrast with the cases where reversals are absent.

Finally, when both Figures 1 and 2 are considered the results of two independently conducted simulation studies of the LDA biplot classification procedure can be compared and similarly for the FDA (MARS) biplot procedure. These comparisons show very similar results for comparable cases (e.g. median values of 70.5 and 72 for the LDA procedure found with Configuration 2; median values of 71.25 and 73 for the FDA (MARS) biplot procedure found with Configuration 2). This finding suggests that the number of Monte Carlo repetitions was adequate.

6. Discussion and conclusions

Possible reasons for the difference in the results obtained from Configurations 2 and 5 on the one hand and Configurations 3 and 6 are suggested when the associated log likelihood ratios are inspected. Keeping in mind that a response pattern in this study consists of six integers each of which can either be zero or one, the respective log likelihood ratios can be considered as functions of the number of ones. In Figure 3 the respective log likelihood ratios are plotted against the number of ones in a response pattern.

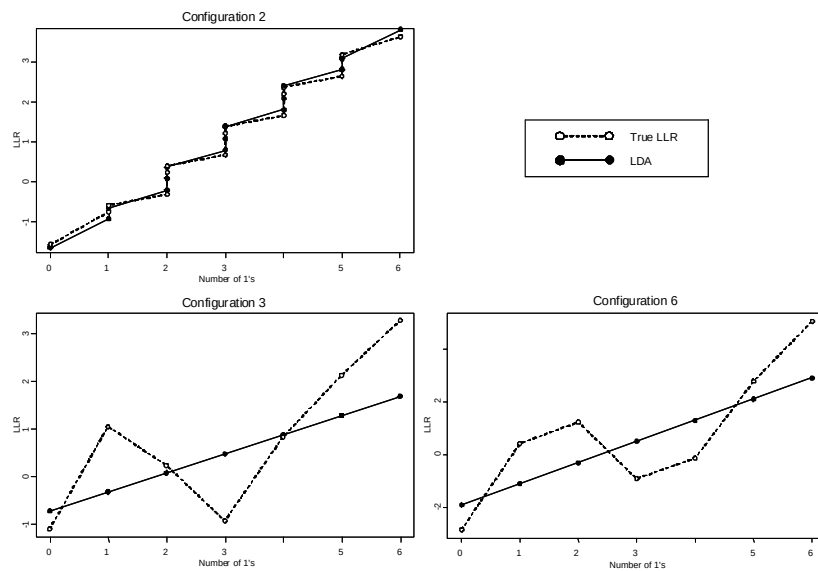


Figure 3. Log likelihood ratios (LLR) of the second order Bahadur model and the LDA classification procedure under Moore Configuration 2 (upper panel); Configuration 3 (bottom left panel) and Configuration 6 (bottom right panel). Each LLR is expressed as a function of the number of ones in a response pattern.

It is clear from Figure 3 that the true log likelihood ratio (LLR) is a non-decreasing function of the number of ones in a response pattern originating from Configuration 2. The log likelihood ratio associated with the LDA biplot procedure follows the true log likelihood ratio very closely. Since the log likelihood ratios in the case of Configuration 5 give a very similar graph to that in the upper panel in Figure 3 it is not included here. In these cases the LDA biplot procedure could be expected to perform well. The FDA biplot procedure having also the advantage of incorporating categorical variables naturally can be considered a non-parametric procedure and performs equally well.

Contrary to the LLR graphs associated with Configurations 2 and 5 the true LLRs in the bottom panels of Figure 3 both show a reversal: negative at first and increasing with the number of ones in the response pattern to a positive value, then decreasing to a negative value again before finally going positive with increasing number of ones in the response pattern. The exact shape of the reversal corresponds to the difference $\pi_2(\underline{x}) - \pi_1(\underline{x})$.

Since the LLR of the LDA procedure in Configurations 3 and 6 is a linear increasing function of the number of ones in a response pattern, the reversal cannot be adequately detected by the LDA biplot procedure resulting in a poor classification performance. The FDA biplot procedure, being essentially a non-parametric procedure, is able to follow the reversal resulting in a significantly better classification performance than the LDA procedure and where comparisons are possible a performance not worse than any of the other procedures considered by Moore (1973). Moreover, it is clear from our results that BRUTO outperforms MARS in cases where no reversal is present. The opposite holds in the presence of reversals. Keeping in mind that BRUTO is

an additive procedure while MARS also takes interactions into account, might point to a possible explanation for the difference in the classification behaviour of the non-parametric (FDA) biplot based classification procedures based upon BRUTO and MARS respectively: the additive BRUTO procedure has the smaller CV error rate in the absence of reversals with the MARS procedure performing better in the presence of reversals. Overparameterisation by the MARS procedure which includes interaction terms in the absence of reversals might be the cause of its inferior performance in such cases.

A large collection of biplots associated with the above simulation study can be constructed. As examples of these biplots a CVA biplot and an FDA (MARS) biplot associated with Configuration 6 for $n_1 = n_2 = 50$ are given in the left and right panels of Figure 4, respectively. These biplots have been equipped with classification regions.

Note that the biplots in Figure 4 are one-dimensional due to the canonical space being one-dimensional for a two-class discriminant analysis. In order to compare misclassified points in these figures, points falling on top of each other have been spread vertically. The improved classification generally found for Configuration 6 when an FDA biplot procedure is being used can now be visually appreciated.

The generalised biplot itself can also be plotted to provide a two-dimensional display of the ordination for obtaining the FDA biplot. An example of such a biplot is given in Figure 5.

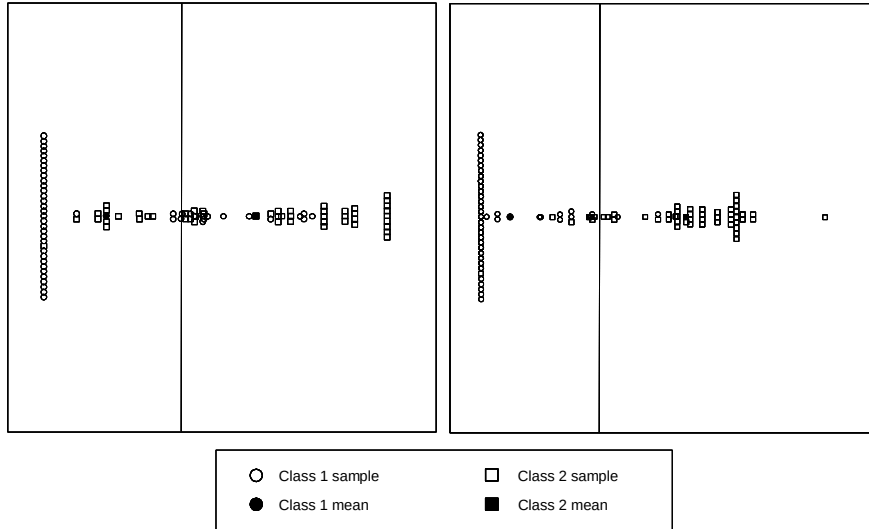


Figure 4. Left panel: CVA biplot of a repetition with $n_1 = n_2 = 50$ of Moore Configuration 6, based on a generalised biplot ordination. Right panel: FDA (MARS) biplot of a repetition with $n_1 = n_2 = 50$ of Moore Configuration 6, based on a generalised biplot ordination. In both biplots the classification regions of classifying a sample into one of the two classes are indicated by a vertical line. Samples are classified according to its distance from the nearest class mean.

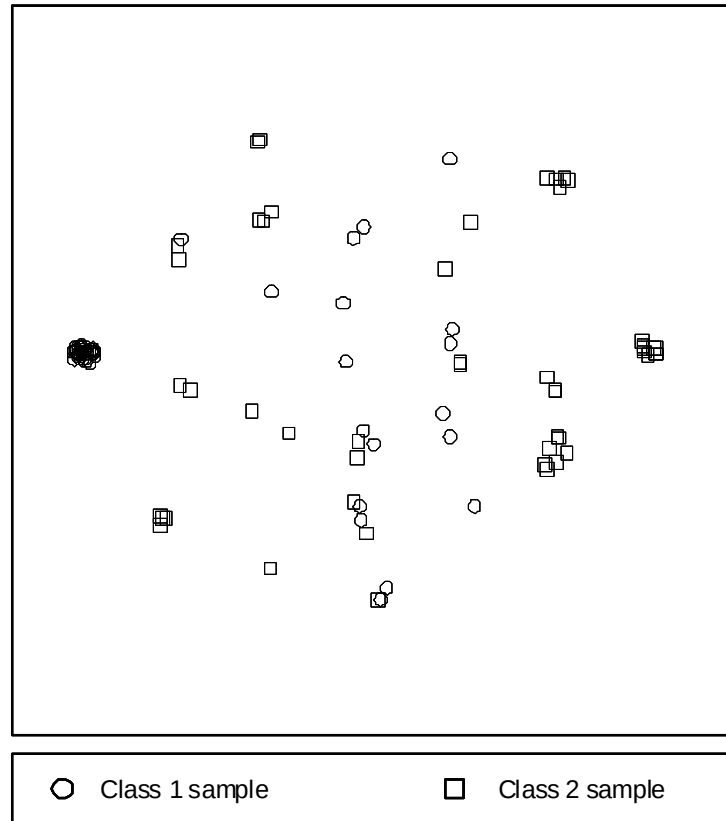


Figure 5. Example of a generalised biplot ordination for a repetition with $n_1 = n_2 = 50$ of Moore Configuration 6.

In conclusion, in this paper

- Discriminant analysis with categorical variables was considered.
- It was shown that biplot methodology lends itself to incorporate categorical variables in a natural way.
- It was demonstrated that the FDA procedures based on biplot methodology performed excellently in case of reversals where it is well known that LDA fails.

- In situations that are favourable for LDA it was found that the FDA procedures gave comparable results.

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